

Advanced Agentic AI:

RAG 2.0, SLMs & Multi-Agent Systems

Course Overview

This intensive 2-day workshop is designed for elite engineers who have already mastered the basics of LLM orchestration and are ready to move to the frontier of Agentic AI. The curriculum covers the "RAG 2.0" stack, moving beyond simple vector search into GraphRAG and recursive retrieval (RAPTOR). Participants will explore the trade-offs of Small Language Models (SLMs), build sophisticated multi-agent systems using LangGraph and CrewAI, and implement production-grade observability and evaluation frameworks.

Note: This is a high-level engineering workshop focusing on the most recent advancements in the GenAI ecosystem (2024-2025 research).

Objectives

- Master advanced prompting techniques and agentic reasoning loops.
- Understand the deployment and optimization of Small Language Models (SLMs).
- Implement state-of-the-art RAG architectures (GraphRAG, HippoRAG, and RAPTOR).
- Architect multi-agent systems using LangGraph and CrewAI with A2A (Agent-to-Agent) protocols.
- Deploy comprehensive observability, prompt management, and evaluation pipelines using Langfuse and DeepEval.
- Explore high-speed agent execution with Agent Lightning.

Duration

2 Days (8 hours/day)

Prerequisites

- Mastery of Python and standard RAG architectures (Vector DBs + LangChain).
- Familiarity with Graph Databases (concepts) and LLM Fine-tuning basics.
- **MANDATORY:** Environment setup (Docker, Python 3.11+, and specific GraphRAG/Ollama dependencies) must be completed prior to Day 1.

Target Audience

- Senior Software Engineers and AI Architects building complex, autonomous systems.
- Lead Engineers responsible for the "AI Ops" stack (Observability & Evals).

Course Outline

Day 1 – Advanced RAG, SLMs & Fine-Tuning (8 Hours)

Module 1: Beyond Simple RAG: The RAG 2.0 Stack

- The limits of "Chunk + Embed" for complex reasoning.
- **GraphRAG:** Leveraging Knowledge Graphs for global document understanding.
- **HippoRAG:** Associative memory and brain-inspired long-term retrieval.
- **RAPTOR:** Recursive Abstractive Processing for Tree-Organized Retrieval.
- **Real-World Example / Hands-on Lab:** Implementing a RAPTOR-based retrieval pipeline to answer "big picture" questions across thousands of unstructured technical documents.

Module 2: Small Language Models (SLMs) & Fine-Tuning

- Why SLMs? (Microsoft Phi-3, Mistral 7B) for edge and cost-efficiency.
- Fine-Tuning LLMs/SLMs: When to tune weights vs. when to use RAG.
- Optimizing SLMs for specific tasks (Summarization, Tool Calling).
- **Real-World Example / Hands-on Lab:** Quantizing an SLM and running it locally using Ollama, then benchmarking its performance against a larger frontier model for a specific Nutanix-related task.

Module 3: Advanced Prompting & Text Generation Apps

- Prompt Engineering at Scale: Self-Correction, Least-to-Most, and DSPy (Programmatic Prompting).
- Building production-grade Text Generation and Chat applications.
- Managing state and long-term memory in complex chat interfaces.
- **Real-World Example / Hands-on Lab:** Creating a DSPy program that automatically optimizes prompts for a multi-step text generation task without manual prompt tweaking.

Day 2 – Agentic AI, Protocols & Observability (8 Hours)

Module 4: Advanced Multi-Agent Frameworks

- Moving from sequential chains to Cyclic Graphs with **LangGraph**.
- Collaborative task management with **CrewAI**.
- Designing specialized agent roles (Architect, Coder, Reviewer).
- **Real-World Example / Hands-on Lab:** Building a multi-agent "Research & Development" crew where a LangGraph agent manages the state and loops between three CrewAI agents to solve a coding bug.

Module 5: Agent Protocols & High-Speed Execution

- **A2A Protocol:** Wrapping agents for standardized communication.
- **Agent Lightning:** Architectures for high-frequency, low-latency agent execution.
- Agent Reinforcement: Feedback loops and agent self-improvement patterns.
- **Real-World Example / Hands-on Lab:** Implementing a standardized A2A interface between two disparate agentic systems to share context and solve a cross-domain infrastructure problem.

Module 6: Observability, Tracing & Prompt Management

- Managing the "Black Box": Introduction to **Langfuse**.
- Distributed tracing for multi-agent loops.
- Versioning and deploying prompts as code.
- **Real-World Example / Hands-on Lab:** Instrumenting the LangGraph agent from Module 4 with Langfuse to visualize token cost, latency, and reasoning steps in real-time.

Module 7: The "Evals" Stack (AgentEvals & DeepEval)

- Why "vibe checks" fail: Introduction to Metric-based Evaluations.
- LLM-as-a-Judge: Using AgentEvals and DeepEval frameworks.
- Testing for hallucinations, faithfulness, and relevancy in agentic outputs.
- **Real-World Example / Hands-on Lab:** Running a "Red-Teaming" suite against your agents using DeepEval to find weaknesses in their reasoning and data retrieval.

Module 8: Course Wrap-up & Future of Agents

- The roadmap to Autonomous AI at Nutanix.

- Security, Ethics, and the "Agentic Future."
- **Final Project Showcase & Q&A.**